

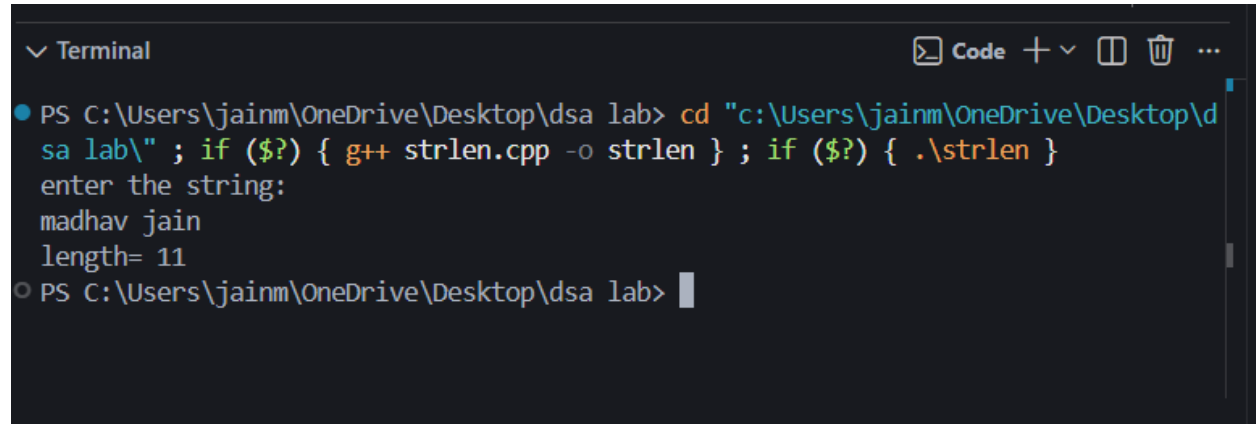
Assignment 3

Q1: Write a program to implement strlen() function.

Code: Without Inbuilt function:

```
#include <iostream>
using namespace std;
int main(){
    char str[100];
    int length=0;
    cout<<"enter the string: "<<endl;
    cin.getline(str,100);
    while (str[length]!='\0'){
        length++;
    }
    cout<<"length= "<<length;
    return 0;
}
```

Output:



```
Terminal
Code + - [ ] [ ] ...
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ strlen.cpp -o strlen } ; if ($?) { .\strlen }
enter the string:
madhav jain
length= 11
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Code: With inbuilt function:

```
#include <iostream>
#include <cstring>
using namespace std;
int main(){
    char str[100];
    cout<<"enter the string: "<<endl;
    cin.getline(str,100);

    cout<<"length= "<<strlen(str);
}
```

```
    return 0;
}
```

Output: With inbuilt function:



```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
enter the string:
Madhav Jain
length= 11
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Q2: Write a program to implement strcpy() function.

Code: Without Inbuilt Function

```
#include <iostream>
using namespace std;
int main(){
    char alpha[100];
    char beta[100];
    cout<<"enter string alpha"<<endl;
    cin.getline(alpha,100);
    int i=0;
    while(alpha[i]!='\0'){
        beta[i]=alpha[i];
        i++;
    }
    beta[i]='\0';
    cout<<"copied string: "<<beta;

    return 0;
}
```

Output: Without Inbuilt function:

```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ strcpy.cpp -o strcpy } ; if ($?) { .\strcpy }
enter string alpha
Madhav Jain
copied string: Madhav Jain
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Code: With Inbuilt function:

```
#include <iostream>
#include <cstring>
using namespace std;
int main(){
    char alpha[100];
    char beta[100];
    cout<<"enter string alpha"<<endl;
    cin.getline(alpha,100);
    strcpy(beta,alpha);
    cout<<beta;

    return 0;
}
```

Output: With Inbuilt function:

```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
enter string alpha
Madhav Jain
Madhav Jain
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Q3: Write a program to implement strcat() function.

Code: Without Inbuilt function

```
#include <iostream>
using namespace std;
int main(){
    char str1[100];
    char str2[100];
    cout<<"enter string1: "<<endl;
    cin.getline(str1,100);

    cout<<"enter string2: "<<endl;
    cin.getline(str2,100);
    int i=0;
    while(str1[i] !='\0'){
        i++;
    }
    int j=0;
    while(str2[j] !='\0'){
        str1[i]=str2[j];
        i++;
        j++;
    }
    str1[i]='\0';
    cout<<"concatenated string= "<<str1;

    return 0;
}
```

Output: Without Inbuilt function:

```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ strcat.cpp -o strcat } ; if ($?) { .\strcat }
enter string1:
Madhav
enter string2:
Jain
concatenated string= MadhavJain
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> 
```

Code: With Inbuilt function:

```
#include <iostream>
#include <cstring>
using namespace std;
int main(){
    char str1[100];
    char str2[100];
    cout<<"enter string1: "<<endl;
    cin.getline(str1,100);

    cout<<"enter string2: "<<endl;
    cin.getline(str2,100);
    strcat(str1,str2);
    cout<<str1;

    return 0;
}
```

Output: With inbuilt function:



```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
enter string1:
Madhav
enter string2:
Jain
MadhavJain
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Q4: Write a program to implement strcmp() function.

Code: Without Inbuilt function:

```
#include <iostream>
using namespace std;
int main(){
    char str1[100];
    char str2[100];

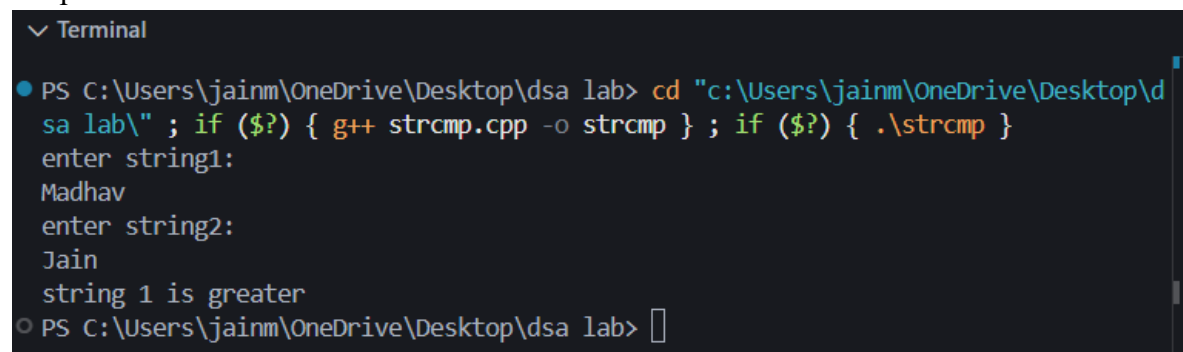
    cout<<"enter string1: "<<endl;
    cin.getline(str1,100);

    cout<<"enter string2: "<<endl;
    cin.getline(str2,100);

    int i=0;
    while(str1[i]==str2[i] && str1[i]!='\0'){
        i++;
    }
    if(str1[i]==str2[i]){
        cout<<"equal strings"<<endl;
    }
    else if(str1[i]>str2[i]){
        cout<<"string 1 is greater"<<endl;
    }
    else{
        cout<<"String 2 is greater"<<endl;
    }

    return 0;
}
```

Output: Without Inbuilt function

A terminal window with a dark background and light blue text. The prompt is 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'. The user enters 'cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if (\$?) { g++ strcmp.cpp -o strcmp } ; if (\$?) { .\strcmp }'. The prompt changes to 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'. The user enters 'enter string1:'. The prompt changes to 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'. The user enters 'Madhav'. The prompt changes to 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'. The user enters 'enter string2:'. The prompt changes to 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'. The user enters 'Jain'. The prompt changes to 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'. The user enters 'string 1 is greater'. The prompt changes to 'PS C:\Users\jainm\OneDrive\Desktop\dsa lab>'.

```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ strcmp.cpp -o strcmp } ; if ($?) { .\strcmp }
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
enter string1:
Madhav
enter string2:
Jain
string 1 is greater
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Code: With Inbuilt function:

```
#include <iostream>
#include <cstring>
using namespace std;
int main(){
    char str1[100];
    char str2[100];

    cout<<"enter string1: "<<endl;
    cin.getline(str1,100);

    cout<<"enter string2: "<<endl;
    cin.getline(str2,100);

    int result=strcmp(str1,str2);
    cout<<result<<endl;

    return 0;
}
```

Output: With Inbuilt function



```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
enter string1:
Madhav
enter string2:
Jain
1
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```

Q5: WAP to demonstrate limitations of Two-Dimensional Array of Characters.

Code:

```
#include <iostream>
using namespace std;
int main(){
    char menu[3][20];
    cout<<"Enter the name of 3 dishes:\n";
    for(int i=0;i<3;i++){
        cin.getline(menu[i],20);
    }
    cout<<"Dishes ordered are: "<<endl;
    for(int i=0;i<3;i++){
        cout<<menu[i]<<endl;
    }

    return 0;
}
```

Output:



```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ chararray.cpp -o chararray } ; if ($?) { .\chararray }
Enter the name of 3 dishes:
Dosa
Chana Bhatura
Soup
Dishes ordered are:
Dosa
Chana Bhatura

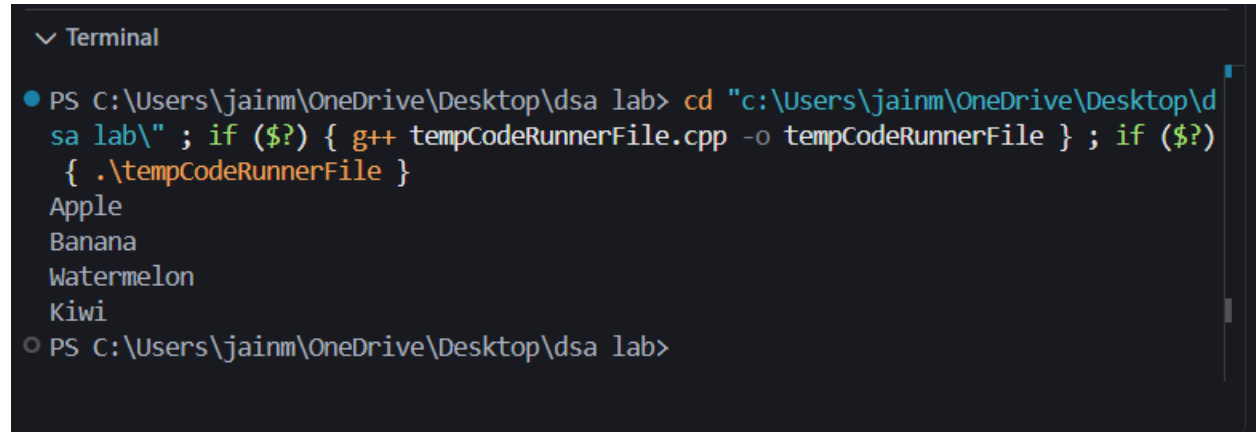
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ chararray.cpp -o chararray } ; if ($?) { .\chararray }
Chana Bhatura
Soup
Dishes ordered are:
Dosa
Chana Bhatura
Soup
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```


Q6: WAP to demonstrate an array of Pointers to Strings.

Code:

```
#include <iostream>
using namespace std;
int main() {
    const char* names[] = {
        "Apple",
        "Banana",
        "Watermelon",
        "Kiwi"
    };
    for (int i = 0; i < 4; i++) {
        cout << names[i] << endl;
    }
    return 0;
}
```

Output:



```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Apple
Banana
Watermelon
Kiwi
PS C:\Users\jainm\OneDrive\Desktop\dsa lab>
```